

Reconstructing 3D chromosome structures from single-cell Hi-C data with SO(3)-equivariant graph neural networks

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Abstract

The spatial conformation of chromosomes and genomes of single cells is relevant to cellular function and useful for elucidating the mechanism underlying gene expression and genome methylation. The chromosomal contacts (i.e. chromosomal regions in spatial proximity) entailing the three-dimensional (3D) structure of the genome of a single cell can be obtained by single-cell chromosome conformation capture techniques, such as single-cell Hi-C (ScHi-C). However, due to the sparsity of chromosomal contacts in ScHi-C data, it is still challenging for traditional 3D conformation optimization methods to reconstruct the 3D chromosome structures from ScHi-C data. Here, we present a machine learning-based method based on a novel SO(3)-equivariant graph neural network (HiCEGNN) to reconstruct 3D structures of chromosomes of single cells from ScHi-C data. HiCEGNN consistently outperforms both the traditional optimization methods and the only other deep learning method across diverse cells, different structural resolutions, and different noise levels of the data. Moreover, HiCEGNN is robust against the noise in the ScHi-C data.

Introduction

The three-dimensional (3D) structure of chromosomes and genomes is an important determinant of cellular function and is involved in many biological processes such as gene regulation [1–3]. Research has shown that the spatial organization of chromosomes is intimately linked to gene expression patterns, with specific genes and regulatory elements adopting distinct positions within the 3D nuclear space [4, 5]. This organization is pivotal in orchestrating cellular differentiation and development, ensuring precise DNA replication and repair, and maintaining cellular identity [6, 7]. Epigenetic modifications, such as DNA methylation and histone modifications, further contribute to the dynamic nature of chromosomal architecture, impacting gene accessibility and function [8].

The disruption of chromosomal structure has been implicated in various diseases, including cancer and genetic disorders [9]. Investigating the 3D chromosome structures can enhance our understanding of fundamental biological mechanisms in cells, offering insights into the molecular basis of diseases and potential avenues for therapeutic interventions.

Advanced experimental techniques 3C-based methods like Hi-C [5, 10–12] have provided valuable insights into the genome's spatial organization, revolutionizing our understanding of the functional implications of chromosomal 3D structure. They apply high-throughput sequencing techniques to sequence pairs of DNA sequence reads that are in spatial proximity (i.e. chromosomal contacts), which have enabled scientists to probe the spatial organization of chromosomes within the nucleus with unprecedented resolution. It has led to the emergence of computational algorithms and modeling

techniques to reconstruct the 3D structures of chromosomes and genomes from chromosomal contact data derived from Hi-C data [13–28]. For a detailed review of the constrained optimization methods for 3D chromosome and genome structure modeling, readers are referred to [29].

Most modeling methods adopt a constrained optimization strategy of searching for a 3D structure that satisfies the distances or contacts between chromosomal regions as well as possible, where the expected distances between two chromosomal regions (also called “wish” distances) are computed from the number of contacts between them (i.e. contact frequency) in the Hi-C data through an inverse power law function [5]. Then, starting from an initial structure of a chromosome or genome in which the positions (i.e. X, Y, and Z coordinates) of chromosomal regions (also called chromosomal bins) are (largely) randomly initialized, they optimize the X, Y, and Z coordinates of the chromosomal regions to make the Euclidean distance between them in the 3D structure as close as possible to the “wish” distance between them, using various optimization algorithms, such as the constrained nonconvex optimization [14], IPOPT optimization solver (open-source software) [26], Integrated Modeling Platform (IMP) [30], Markov chain Monte Carlo (MCMC) [20], Poisson regression approach [16], gradient descent optimization [21, 22], semi-definite programming [27], the shortest path algorithm [17], and low-rank matrix completion techniques [25].

Despite the significant progress, the distance-based optimization methods have some limitations. First, the same optimization method (e.g. MCMC simulation or gradient descent

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simulation) is applied to every Hi-C dataset (e.g. chromosome distance matrices) to build chromosome/genome structures, which is very slow when the distance matrix is very large. The problem is particularly severe for time-consuming optimization methods, such as MCMC. Also some of the methods have many optimization parameters to adjust [21, 22], which are difficult for users to use. Second, because the same optimization process is used on every Hi-C dataset separately, the optimization methods cannot learn common chromosomal contact and structural patterns across different Hi-C datasets, resulting in sub-optimal performance. Third, most of the optimization methods primarily concentrate on reconstructing 3D structures from bulk Hi-C data of a population of cells, characterized by dense chromosomal contacts, which do not work well for single-cell Hi-C (ScHi-C) data that contain much fewer chromosomal contacts (e.g. <5% of chromosomal contacts in bulk Hi-C data).

A totally different approach to generating 3D chromosome structures from Hi-C data is the machine learning approach. To date, only one machine learning method, Hi-C data Graph Neural Network (HiC-GNN) [15], utilizing a graph convolutional neural network method, has been developed to directly generate the 3D structure from bulk Hi-C data of a population of cells, demonstrating the machine learning approach is viable for the 3D chromosome modeling task. The machine learning approach has some advantages. First, using the pretrained machine learning model can be much faster and simpler than using the traditional optimization methods to generate chromosome/genome structures from Hi-C. Second, because the machine learning models are trained on large amounts of ScHi-C data, they can learn common patterns in the input data and output structures that the traditional optimization methods applied to each dataset separately may not be able to capture. Similar advantages of using deep learning methods have already been successfully shown in other studies, such as [31, 32]. In this work, we developed a new machine learning method based on a novel SO(3)-equivariant graph neural network (called HiCEGNN) to reconstruct 3D chromosome structures from ScHi-C data, addressing the limitations of the existing methods mentioned above. Our approach uses a graph to represent chromosomal contact matrices derived from ScHi-C data, where each region (bin) of a chromosome is considered a node, and a pair of nodes with contact frequency >0 is connected by an edge. The graph for a chromosome is fed into a SO(3)-equivariant graph neural network (EGNN) that is equivariant to all possible rotations in the 3D space (i.e. the special orthogonal group in three dimensions) to predict the X, Y, and Z coordinates of each node, i.e. the structure of the chromosome. Benchmarked on several ScHi-C datasets, HiCEGNN outperforms three traditional optimization methods (ShRec3D [17], 3DMax [19], and ShNeigh [33]) and the other deep learning method—HiC-GNN [15]—across various cells, different resolutions, and different noise levels. Moreover, it can also reliably reconstruct chromosome structures from bulk Hi-C datasets.

Materials and methods

An overview of HiCEGNN

The chromosome structure prediction pipeline of HiCEGNN is outlined in Fig. 1A, which consists of two main parts: (i) a ScHi-C data processing block and (ii) a SO(3)-EGNN block.

The input for the data processing block is a normalized ScHi-C contact matrix of a chromosome, which contains contact frequencies between chromosomal bins. This matrix is processed through three parallel operations (Fig. 1A) to construct a graph representation consisting of node features, edge features, and the initial coordinates of the nodes. The first operation is to embed the ScHi-C data matrix into “node features” describing chromosomal bins using the LINE algorithm [34], which is designed specifically for generating node features for large graphs and has been successfully validated on Hi-C data [35]. The second operation is to use nonzero contact frequency between two chromosomal nodes (bins) as the “features of the edges” connecting them. The third operation is to convert the ScHi-C contact matrix into a distance matrix based on an inverse power-law relationship between distance and contact frequency [5]. The “initial coordinates” for the nodes are then estimated using the eigenvalue decomposition of the Gram matrix [36], which is derived from the corresponding distance matrix.

The three data processing operations above produce a graph representation for each ScHi-C contact matrix, in which each node i has initial features b_i and initialized coordinate $x_i \in X$ and each edge has features e_{ij} . The graph is passed to an SO(3)-EGNN with 4 graph neural network (GNN) layers to predict the 3D structure of the chromosome. The equivariance of EGNN means if the initial 3D input structure is rotated or translated, the same hidden features can be generated at the corresponding rotated/translated position in the hidden space, which is important for the neural network to recognize the patterns in the same 3D object in different orientations or at different locations.

The detailed architecture of the EGNN is shown in Fig. 1B. It first centralizes all the nodes in the graph based on the initial coordinates [37] by making the centroid of all the nodes in a many-body system (i.e. the graph) as the origin to ensure the coordinates are translation-equivariant. The centralized coordinates and edge features form a local equivariant complete frame F_{ij} [37] to avoid direction degradation caused by any force direction in a many-body system. Then, the position x_i of each node can be projected onto the complete frame to get corresponding scalarization coefficients t_{ij} . Then all the scalarization coefficients t_{ij} , the edge features e_{ij} , and the node features b_i are fed to the graph message-passing block that is repeated four times to produce the final 3D chromosome structure. The loss function of HiCEGNN is the mean squared difference between the Euclidean distance between any two nodes in the 3D structure and their “wish” distance calculated from their interaction frequency. HiCEGNN was trained on ScHi-C datasets from human frontal cortex with 24 chromosomes (Chromosomes 1–22, X, and Y) [38] (GEO accession number: GSE130711) to generate 3D genome structures by minimizing the loss.

After being trained, HiCEGNN was blindly tested on multiple new ScHi-C datasets of various resolution from the same cell line as well as different cell lines. It was also tested on bulk Hi-C data to check if it could be applied to them. All the test datasets consist of the Hi-C data of the chromosomes that are excluded from both the training and validation datasets to guarantee the objective assessment of the generalization capability of HiCEGNN.

The datasets, data preprocessing, and methods of HiCEGNN are described in detail in the sub-sections below.

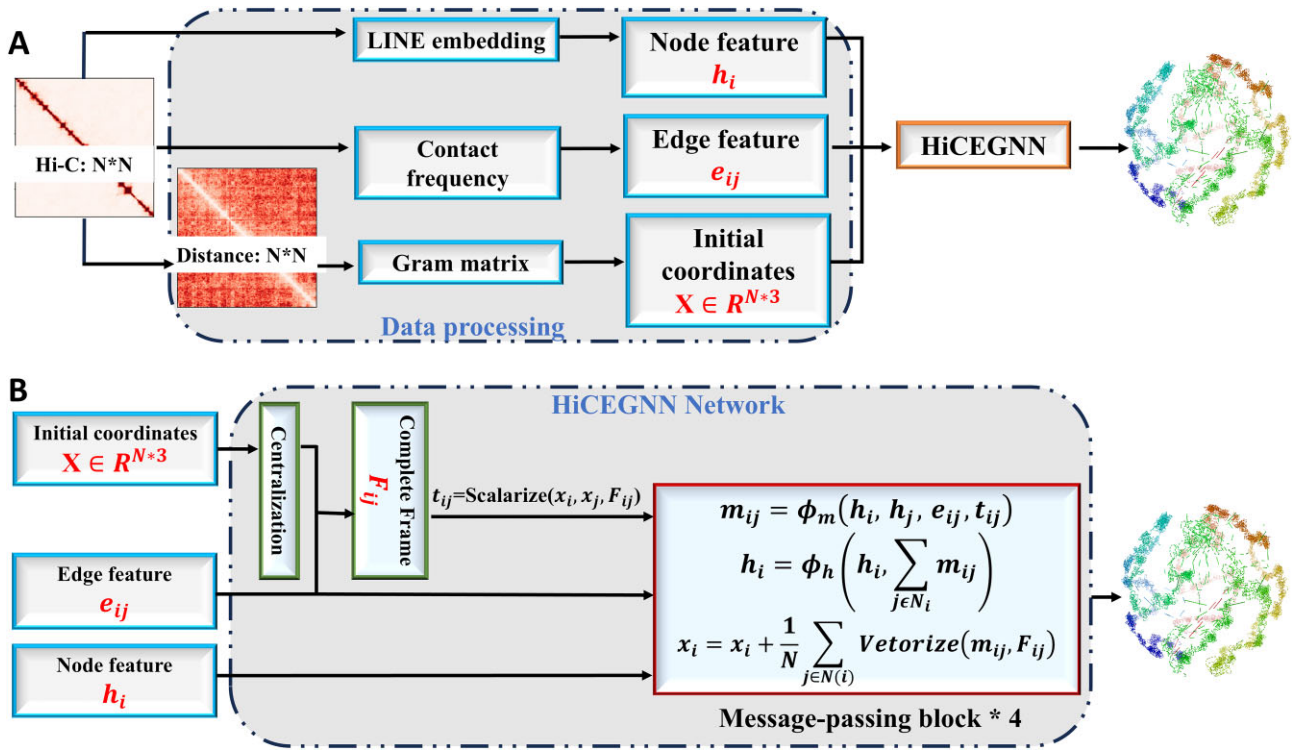


Figure 1. The over architecture of HiCEGNN. (A) A normalized Hi-C contact matrix (input) of a chromosome is processed to construct a graph representation for HiCEGNN to generate the 3D structure of the chromosome. (B) The architecture of the SO(3)-EGNN of HiCEGNN to predict the 3D chromosome structure.

Hi-C datasets

The single-cell and bulk Hi-C datasets of two cell lines [human frontal cortex cell and *Drosophila* male Dm-BG3c2 (BG3) cell] used in this work are obtained from a previous study [39]. *human_cell_1_training_data* contains the ScHi-C data of Chromosomes 1, 3, 5, 7, 8, 9, 11, 13, 15, 16, 17, 19, 21, and 22 from a single cell (called human cell 1), which was used to train HiCEGNN to predict 3D chromosome structures from ScHi-C data. *human_cell_1_validation_data* contains the ScHi-C data of Chromosomes 2, 6, 10, and 12 from the human cell 1, which was used to monitor the performance of HiCEGNN during training and select a trained model for blind test. *human_cell_1_test_data* consists of the ScHi-C data of Chromosomes 4, 14, 18, and 20 from human cell 1, which was used to blindly evaluate HiCEGNN's performance on chromosomes not used in training and validation. *human_cells_2_3_test_data* comprises the ScHi-C data of two randomly selected human cells (human cell 2 and human cell 3), which was used to blindly test how well HiCEGNN trained on human cell 1 worked on different human cells. *drosophila_cells_test_data* contains the ScHi-C data of chromosomes chr2L, chr2R, chr3L, chr3R, chr4, chrX, and chrM of two randomly selected *Drosophila* cells, which was used to blindly test how well HiCEGNN trained on human cell 1 worked on the ScHi-C data of another species.

In addition to the ScHi-C datasets above, the following bulk datasets were also used to train and test HiCEGNN. *human_population_training_data* comprises bulk Hi-C data of Chromosomes 1, 3, 5, 7, 8, 9, 11, 13, 15, 16, 17, 19, 21, and 22 from a population of the human cells, which was used to train HiCEGNN to predict 3D chromosome structures from bulk Hi-C data. *human_population_validation_data* com-

prises the bulk Hi-C data of Chromosomes 2, 6, 10, and 12 from a population of the human cells, which was used to monitor the performance of HiCEGNN on bulk Hi-C data during training and select a trained model for blind test. *human_population_test_data* consists of the bulk Hi-C data of Chromosomes 4, 14, 18, and 20 from a population of human cells, which was used to blindly evaluate HiCEGNN's performance on bulk Hi-C data.

Data preprocessing, graph representation, and feature generation

The Hi-C reads at several specific representative resolutions (i.e. 40, 80, 160, and 320 kb) for each chromosome from the human ScHi-C dataset were first extracted from the ".mcool" files of the dataset comprising chromosomal contact reads pre-computed at the following specific resolutions: 10, 20, 40, 80, 160, 320, 640, 1 280, 2 560, 5 120, and 10 240 kb. The upper resolution limit was set to 40 kb because all the methods tested in this study could not build good chromosome structures for higher resolutions (10 kb or 20 kb) for this dataset, probably because the chromosome contact data at those resolutions are too sparse for them to deal with. The extracted reads were used to construct a contact matrix for each chromosome. The interaction frequencies (contact numbers between chromosomal bin of a chromosome) were normalized into the range [0, 1] according to [40], which are represented by a 2D matrix (see a visualized example in Fig. 1A). The normalized matrix M is an adjacency matrix defining a graph, in which each nonzero contact number ($M[i,j]$) between two bins (corresponding to two nodes in the graph) i and j specifies an edge connecting two nodes. The contact number of an edge is

considered as its features (called edge features). Let L denote the total number of edges. The shape of the edge features is $L \times 1$.

A graph embedding method, LINE [34], which had been used to predict chromosome compartmentalization from Hi-C data [35], is employed to generate node features with a dimension of 512 from a normalized contact matrix M . Previous studies, such as HiC-GNN [15], have demonstrated that node feature embeddings with 512 dimensions yield optimal performance. Therefore, we adopted the same embedding dimensions for node features in our model. Let N denote the number of nodes for a chromosome. The shape of the node features is $N \times 512$.

The initial positions (X , Y , and Z coordinates) of the nodes are estimated from the Gram matrix ($\tilde{D}$) calculated from the distances between chromosomal bins through a series of transformations below.

First, the distance map storing the Euclidean distances between two nodes is calculated from the normalized chromosomal contact matrix by Equation (1) according to the inverse power-law function that works well on bulk Hi-C data [5]:

$$D_{ij} = \left(\frac{1}{F_{ij}} \right)^\gamma, \quad (1)$$

where D_{ij} is the distance between two bins (nodes) i and j , F_{ij} is the contact number between the two nodes, and γ is the conversion factor set to 0.5.

Second, The Gram matrix ($\tilde{D}$) is computed from the distance map D by Equation (2):

$$\tilde{D}_{ij} = 0.5 (D_{i,1}^2 + D_{1,j}^2 - D_{i,j}^2). \quad (2)$$

Third, an eigenvalue decomposition [36] is applied to $\tilde{D}$ according to Equation (3):

$$\tilde{D} = USU^T, \quad (3)$$

here S is a diagonal matrix containing all eigenvalues in the descending order.

Finally, the X , Y , and Z coordinate of each node i (denoted as x_i) is estimated as the first three columns of row i in X , which is computed according to Equation (4):

$$x_i = [X_{i,1}, X_{i,2}, X_{i,3}], X = U\sqrt{S}. \quad (4)$$

The data processing above produces a graph representation of the Hi-C data of a chromosome consisting of node features, edge features, and the initial estimated position (coordinates) of each node, which are the input for HiCEGNN to predict the final 3D chromosome structure. The same data processing is applied to all the single-cell or bulk Hi-C datasets.

The architecture of HiCEGNN

The architecture of HiCEGNN is illustrated in Fig. 1B. It contains the following main components: (i) centralization to ensure many-body system translation-equivariance; (ii) complete frame to formulate a local orthonormal basis of SO(3)-equivariant frame; (iii) scalarize operation to convert geometric tensors into edgewise SO(3)-invariant scalar coefficients/features; and (iv) message-passing block to learn the invariant edgewise embeddings by passing and integrating the scalars on graphs, which contains the vectorize operation to translate scalar representation back to tensors. The four components are described in detail as follows.

Centralization:

Given the coordinates of chromosomal bins of a chromosome structure: $X(t) = (x_1(t), x_2(t), \dots, x_n(t))$ at time t , the centroid $C(t)$ is defined by

$$C(t) = \frac{x_1(t) + x_2(t) + \dots + x_n(t)}{n}.$$

Translate the system by S , then

$$X(t) + S \rightarrow C(t) + S.$$

Therefore, recenter the origin of the reference to the centroid at time $t = 0$ by translation S , we have

$$X(t) - C(t) \rightarrow X(0) - C(0).$$

Therefore, the centralized $X(t)$ can make the chromosome structure (a many-body system) translation-equivariant.

Complete frame:

Based on the previous centralized $X(t)$, for each paired chromosomal bin (particle) $(x_i(t), x_j(t))$, let $a_{ij}^t = \frac{x_i(t) - x_j(t)}{\|x_i(t) - x_j(t)\|}$.

Define $b_{ij}^t = \frac{x_i(t) \times x_j(t)}{\|x_i(t) \times x_j(t)\|}$ and $c_{ij}^t = a_{ij}^t \times b_{ij}^t$, then we have the following expression:

$$F_{ij} = (a_{ij}^t, b_{ij}^t, c_{ij}^t).$$

This complete frame F_{ij} is assumed to be a nondegenerative SO(3)-equivariant frame.

Scalarize operation:

Based on the complete frame $F_{ij} = (a_{ij}^t, b_{ij}^t, c_{ij}^t)$, the particle x_i can be projected onto it with the following corresponding coefficients' set, which is a scalar tuple $(x_i \cdot a_{ij}^t, x_i \cdot b_{ij}^t, x_i \cdot c_{ij}^t)$. The scalar tuple is a SO(3)-invariant scalar tuple and the process of obtaining the scalar tuple can be defined as a scalarize operation, which can be expressed by the following equation:

$$\begin{aligned} t_{ij} &= \text{Scalarize}(x_i, x_j, F_{ij}) \\ &= (x_i \cdot a_{ij}^t, x_i \cdot b_{ij}^t, x_i \cdot c_{ij}^t, x_j \cdot a_{ij}^t, x_j \cdot b_{ij}^t, x_j \cdot c_{ij}^t). \end{aligned}$$

This operation could help encode the geometric tensors, such as $X(t)$, into SO(3)-invariant scalars.

Message-passing block:

After we get all the SO(3)-invariant scalars based on the scalarize operation, within the message-passing block, we first embed the scalars t_{ij} that contain the geometric information together with edge features e_{ij} and node features h_i into high dimensional representations according to the following equations:

$$t_{ij} = \text{MLP}(t_{ij}) \text{ or } \text{Fourier}(t_{ij}),$$

$$h_i = \text{MLP}(h_i),$$

$$e_{ij} = \text{MLP}(e_{ij}).$$

Here, MLP denotes a fully connected multi-layer perceptron and Fourier denotes Fourier transformation with learnable parameters.

Then, the message-passing block takes as input the set of node feature embeddings h_i , edge feature embeddings e_{ij} , the set of SO(3)-invariant scalars t_{ij} , and the complete frame F_{ij} calculated from the coordinates X to update the node features

and coordinates by the following equations:

$$m_{ij}^l = \varnothing_m \left(h_i^l, h_j^l, e_{ij}^l, t_{ij}^l \right),$$

$$h_i^{l+1} = \varnothing_h \left(h_i^l, \sum_{j \in N(i)} m_{ij}^l \right),$$

$$F_{ij}^{l+1} = \text{Complete frame} \left(x_i^l, x_j^l \right) = \left(a_{ij}^l, b_{ij}^l, c_{ij}^l \right),$$

$$x_i^{l+1} = x_i^l + \frac{1}{N} \sum_{j \in N(i)} \text{Vectorize} \left(m_{ij}^l, F_{ij}^l \right).$$

Here, $\varnothing_m$ and $\varnothing_h$ are MLPs with different learnable parameters. Since the message-passing block is processing the invariant scalars t_{ij} , its outputs are still invariant scalars when applied on any nonlinear activation functions. There are four message-passing blocks forming a cascade in the HiCEGNN model.

Training and loss function

The graph representations of chromosomes in *human_cell_1_training_data* (or *human_population_training_data*) at each resolution of 40, 80, 160, and 320 kb were separately used to train HiCEGNN at the resolution to predict chromosome structures from ScHi-C data (or bulk Hi-C data), respectively. During the training, *human_cell_1_validation_data* (or *human_population_validation_data*) at the corresponding resolution was used to monitor its performance to select the final trained HiCEGNN model for blind test on the ScHi-C test data (or the bulk Hi-C test data) at the corresponding resolution.

The output of HiCEGNN is the predicted X, Y, and Z coordinates of all the nodes (chromosomal bins) that define a 3D chromosome structure. The coordinates are converted into a predicted distance map (or matrix) P , in which $P[i, j]$ is the distance between nodes i and j computed from their coordinates. The “wish” distance map (W) for a chromosome, in which $W[i, j]$ is the “wish” distance between nodes i and j , is calculated by Equation (1) and serves as the labels for training HiCEGNN. W has both nonzero values and zero values, where the former represents the meaningful distances between nodes and the latter means unknown/uncertain distances between uncontacted nodes. Therefore, the loss is only calculated for the distances that are nonzero in W and their counterparts in P , which is the mean square difference between the nonzero distances in W and their corresponding predicted distances in P . The loss function of HiCEGNN at each specific resolution was minimized by the Adam optimizer with a learning rate of 0.001 to adjust its weights. After training with different hidden GNN layers, ranging from 1 to 6, the best HiCEGNN model with 4 GNN layers at each specific resolution was selected based on the minimal loss validated on the validation dataset and then was blindly tested on the corresponding test datasets at the specific resolution.

Testing HiCEGNN on test data with additionally added noise

To test whether HiCEGNN is robust against the noise in Hi-C data, we separately added two levels (low and high) of

Gaussian noise to the *human_cell_1_test_data* at a resolution of 320 kb and compared it with the other methods on the noise-added data. According to the previous research [41], the low noise was generated by the normal distribution $N(\delta, \delta)$, where δ is the minimum nonzero value in an input chromosomal contact matrix; and the high noise was generated by the normal distribution $N(3\delta, \delta)$. The two levels of Gaussian noise were added to every nonzero entry in the original chromosomal contact matrices to construct the noisy data, respectively, to create two noise-added datasets. The noise-added datasets are processed in the same way as described in “Section 2.2” to produce the graph representations as input for the pretrained HiCEGNN to predict 3D chromosome structures.

Evaluation metrics

Two metrics, the distance Spearman correlation coefficient (dSCC) and the distance root mean square error (dRMSE), are used to evaluate the 3D chromosome structures constructed by HiCEGNN and other methods as in previous studies [19, 23]. The dSCC is a nonparameterized measure to calculate the rank correlation between the distance map calculated from a predicted structure for a chromosome and the “wish” distance map calculated from the Hi-C contact map of the chromosome, which is described by Equation (5). The dSCC is in the range $[-1, 1]$ and scale-invariant, where a larger value indicates the better quality of the predicted structure:

$$\text{dSCC} = \frac{\sum_{i \in \bar{D}} (m_i - \bar{m}) \sum_{i \in D} (n_i - \bar{n})}{\sqrt{\sum_{i \in \bar{D}} (m_i - \bar{m})^2 \sum_{i \in D} (n_i - \bar{n})^2}}, \quad (5)$$

here, $\bar{D}$ and D are the predicted distance map and the “wish” distance map, respectively. m_i and n_i are the rank of distance i in $\bar{D}$ and D , respectively. $\bar{m}$ and $\bar{n}$ are the mean of all rank in $\bar{D}$ and D , respectively.

dRMSE measures the similarity of the two distance maps using Equation (6). Smaller dRMSE, more similar are the two maps. Different from dSCC, its value is influenced by the absolute value of distances and therefore is not scale-invariant. Therefore, the comparison of dRMSE of the two predicted structures that are at different scales might not be meaningful.

The dRMSE is calculated by Equation (6):

$$\text{dRMSE} = \sqrt{\frac{1}{n} \sum (\bar{d}_{ij} - d_{ij})^2}, \quad (6)$$

here, $\bar{d}_{ij}$ and d_{ij} are the distance between two nodes i and j in $\bar{D}$ and D , respectively.

It is worth noting that, for both Equation (5) and Equation (6), only the distances in the predicted ($\bar{D}$) and wish (D) distance maps corresponding to nonzero entries in the original normalized chromosome contact matrix are used to calculate dSCC and dRMSE.

Results

HiCEGNN reconstructs better 3D chromosome structures from ScHi-C data than existing methods

To benchmark the performance of HiCEGNN on the pre-processed ScHi-C data [38, 42], we first select three traditional optimization methods (3DMax [19], ShNeigh [33], and ShRec3D [17]) that perform well on bulk Hi-C data and the

Table 1. Comparison of five methods on reconstructing 3D chromosome structures from the single-cell Hi-C data of *human_cell_1_test_data*

Resolution	Metrics	Method				
		Traditional optimization method			Deep learning	
		3DMax	ShNeigh	ShRec3D	HiC-GNN	HiCEGNN
320 kb	dSCC ↑	0.3756	0.3555	0.3139	0.2642	0.5728
	dRMSE ↓	8.4724	1.3874	71.6758	0.5814	0.1672

dSCC and dRMSE values for each method are averaged over all the test chromosomes in the dataset. Bold font denotes the best results.

only deep learning model, HiC-GNN [15], for chromosome structure reconstruction to compare with HiCEGNN. For the sake of fairness, all the methods are tested on the same test datasets, and their performance is compared on the same set of chromosomes for which they can build 3D structures.

All the methods were first tested on ScHi-C data of *human_cell_1_test_data* at a relatively low resolution of 320 kb (i.e. 320 000 base pairs per chromosome bin) because all of them can generate 3D structures at this resolution. The dSCC and the dRMSE between the chromosomal distances in the predicted structures and the “wish” distances computed from the normalized chromosomal contact matrices are used to evaluate the methods. The “wish” distances for all the methods are computed by the inverse power law function (Equation 1), whose conversion factor parameter is set to 0.5 as recommended in [25]. [Supplementary Fig. S1](#) shows HiCEGNN performs well across a wide range of the conversion factor values, including 0.5, confirming the commonly used conversion factor value is reasonable.

The results of HiCEGNN and the four other methods on the single-cell data of test chromosomes of human cell 1 are reported in Table 1. HiCEGNN has the highest dSCC value and lowest dRMSE value. The dSCC value of HiCEGNN is 0.5728, much higher than 0.3756 of the second-best method—3DMax—and more than double that of HiC-GNN. Its dRMSE is 0.1672, much lower than all other methods. It is worth noting that because the two deep learning methods, HiCEGNN and HiC-GNN, were trained to minimize the mean square difference between the distances in the 3D structures and the “wish” distances, they have a much lower dRMSE than the nonmachine learning optimization methods, such as 3DMax and ShRec3D that do not directly optimize the mean square difference between the distances. However, among the two deep learning methods, only HiCEGNN performs better than 3DMax and ShRec3D in terms of both dSCC and dRMSE, while HiC-GNN has a worse (lower) dSCC than them. Because dSCC is independent of the scale of distances, it may be a better metric to compare different methods based on the different optimization criteria.

Because all the methods except HiCEGNN are mainly developed to reconstruct 3D structures from bulk Hi-C that contains much more sequence reads (chromosomal contacts) than ScHi-C data. We also compared them on a bulk Hi-C dataset—*human_population_test_data*. Similar results are observed in [Supplementary Table S1](#). HiCEGNN has the highest dSCC and the lowest dRMSE among the five methods. The results show that HiCEGNN can work well on both dense bulk Hi-C data and sparse ScHi-C data, indicating its design is robust.

We also benchmarked HiCEGNN against several other traditional tools designed for reconstructing 3D structures from ScHi-C data, including *nuc_dynamics* [43], *DPDchrom* [44],

SCL [45], and *LJ3D* [46]. After evaluating these tools, we found that only *SCL* and *LJ3D* were still functional. However, because *SCL* and *LJ3D* require a specific format with some information that does not exist in our preprocessed test dataset, they cannot be tested on it. So, we compared HiCEGNN with *SCL* and *LJ3D* on a single-cell test dataset used to benchmark *LJ3D* before. This new test dataset focuses on the X chromosome from TH1 cell 1 and contains 616 intra-chromosomal contacts. We let the X chromosome be represented by 520 beads at a resolution of 320 Kb, a configuration compatible with all the tested methods.

Each traditional nondeep learning method (i.e. *SCL*, *LJ3D*, *ShRec3D*, *ShNeigh*, and *3DMax*) was run on the test dataset five times to generate five structures, while the two deep learning methods (*HiCEGNN* and *HiC-GNN*) were only executed once because they produced the same structure given the same input. As shown in [Supplementary Table S2](#), HiCEGNN substantially outperforms all the other tools, including *SCL*, *LJ3D*, *HiC-GNN*, *ShRec3D*, *ShNeigh*, and *3DMax* in terms of dSCC and dRMSE. We calculated the *P*-value to determine whether the dSCC of HiCEGNN differs significantly from those of traditional methods, including *SCL*, *LJ3D*, *ShRec3D*, *ShNeigh*, and *3DMax*. The average score of the five structures generated by each traditional method is compared with that of HiCEGNN by *t*-test implemented in SciPy Python package [47], yielding a *P*-value of 7.3646e−06 for *3DMax*, 1.7475e−05 for *ShNeigh*, 1.3277e−05 for *ShRec3D*, 3.8379e−05 for *SCL*, and 1.0269e−05 for *LJ3D*. The results demonstrate that HiCEGNN performs significantly better than the other methods.

Hi-C assays are widely utilized to identify chromatin organization features, such as A/B compartments, chromatin loops, and topologically associating domains. Among these, chromatin loops often attract a lot of attention due to their role in regulating genes. However, there were few efforts to identify loops in 3D chromosome structures derived from ScHi-C data. Here, we used *GenomeFlow* [48] to identify putative loops in the reconstructed 3D structures generated by the different tools to compare them. To ensure a fair comparison, we used the five methods (*HiCEGNN*, *HiC-GNN*, *3DMax*, *ShNeigh*, and *ShRec3D*) to reconstruct the 3D structures of chromosome 4, consisting of 588 nodes, from human cell 1, at the same resolution of 320 kb ([Supplementary Fig. S2](#)). Among the five tools, only HiCEGNN and *3DMax* successfully identified some putative loops in the reconstructed 3D structures. HiCEGNN identified four putative loops involving node pairs: 130–373, 379–528, 533–550, and 557–584. Two of these loops are highlighted by black lines (379–528) and green lines (533–550) in the 3D structure ([Supplementary Fig. S2\(A\)](#)). By comparison, *3DMax* identified three putative loops involving node pairs: 3–227, 389–567, and 568–584.

Table 2. Comparison of the five methods on the single-cell Hi-C data of *human cell 2* in *human_cells_2_3_test_data* dataset

Resolution	Metrics	Model				
		Traditional method			Deep learning	
		3DMax	ShNeigh	ShRec3D	HiC-GNN	HiCEGNN
320 kb	dSCC ↑	0.3306	0.3946	0.3135	0.2529	0.4228
	dRMSE ↓	8.5903	1.4759	98.9331	0.5723	0.4454

Bold font denotes the best result.

Table 3. Comparison of the five methods on the single-cell Hi-C data of *Drosophila cell 1* in *drosophila_cells_test_data* dataset

Resolution	Metrics	Model				
		Traditional method			Deep learning	
		3DMax	ShNeigh	ShRec3D	HiC-GNN	HiCEGNN
320 kb	dSCC ↑	0.3922	0.3388	0.2471	0.2352	0.4009
	dRMSE ↓	8.5870	1.2981	118.5976	0.5943	0.2610

Bold font denotes the best result.

It is worth noting that, due to the sparsity of ScHi-C data, directly resolving chromatin loops at the single-cell level remains challenging without aggregating data from multiple cells, even though some studies [49, 50] have provided indirect evidence or probabilistic inferences of loops at the single-cell level. As a result, no ScHi-C experiment has consistently detected chromatin loops with high confidence. Therefore, reliable, direct loop identifications can mostly be made at the population (bulk) cell level. To compare the putative loops predicted by HiCEGNN and 3DMax at the single-cell level with those detected at the bulk cell level, we randomly selected 20 individual human cells and pooled their ScHi-C data together to create a pseudo-bulk Hi-C dataset. We then reconstructed 20 structures of chromosome 4 at the same resolution of 320 kb from the dataset using HiCEGNN and 3DMax, respectively. The 20 structures were then pooled together to generate a consensus structure for each tool. From the pseudo bulk Hi-C data, HiCEGNN identified four putative loops involving the node pairs: 135–370, 376–522, 536–545, and 559–584, while 3DMax identified three putative loops involving the node pairs: 8–220, 372–558, and 563–584. Although the four loops detected from the ScHi-C data by HiCEGNN (i.e. node pairs: 130–373, 379–528, 533–550, and 557–584) are not identical to those detected from the pseudo bulk Hi-C data by HiCEGNN, they are very close to each other with a shift of only several chromosomal bins. The minor shift is likely due to variability in loop positions across individual cells. Moreover, a single-cell loop (node pair: 557–584) detected by HiCEGNN is close to a bulk-cell loop (node pair: 563–584) detected by 3DMax. The results indicate that the putative loops identified in single cells by HiCEGNN are largely consistent with those detected at the pseudo-bulk cell level. Because the loops detected from bulk Hi-C data are much more reliable than those from ScHi-C data, the results validate the biological accuracy of single-cell chromosome structures built by HiCEGNN.

HiCEGNN generalizes well to different cells, cell types, and species

To compare the generalization ability of the methods, we benchmark the five methods on the ScHi-C data of different cells, cell lines, and cell types.

We first compared HiCEGNN pretrained on human cell 1 with HiC-GNN, 3DMax, ShNeigh, and ShRec3D on two unseen test single cells of the same cell line. The results for human cell 2 and for human cell 3 in the *human_cells_2_3_test_data* are shown in Table 2 and [Supplementary Table S3](#), respectively. HiCEGNN still outperforms all other methods in terms of dSCC and dRMSE, even though the dSCC difference between HiCEGNN and other methods is not as large as on the different chromosomes on the same cell (i.e. on *human_cell_1_test_data* in Table 1).

Moreover, we compared HiCEGNN pretrained on human cell 1 with the other methods on the ScHi-C data of another unseen species—*Drosophila* (on *Drosophila* cell 1 in Table 3 and on *Drosophila* cell 2 [Supplementary Table S4](#)), a species evolutionarily distant from human. HiCEGNN still performs best on *Drosophila* cell 1 in terms of both dSCC and dRMSE, and it performs best on *Drosophila* cell 2 in terms of dRMSE and second best in terms of dSCC. The performance of HiCEGNN on *Drosophila* cells is worse than on the human cells 2 and 3 because HiCEGNN was pretrained on human cells and therefore its parameters were not optimized for *Drosophila* cells. It is expected that if HiCEGNN were trained on *Drosophila* cells, it would work even better.

HiCEGNN outperforms other methods in 3D chromosome reconstruction at different resolutions

To test whether HiCEGNN can construct chromosome structures well from ScHi-C data of different resolutions, especially high resolution, we compare it with the other four methods on the *human_cell_1_test_data* at four resolutions: 40, 80, 160, and 320 kb. The data of the different resolutions were extracted from the multiple-resolution cooler files in the dataset prepared by the original authors of the Hi-C data.

The average dSCC and dRMSE scores of the five methods at different resolutions are visualized in Fig. 2. The data points at some high resolutions for ShNeigh and ShRec3D are missing because they could not generate chromosome structure at the resolutions. The dSCC values indicate that HiCEGNN outperforms all other methods across all resolutions, from low (320 kb) to high (40 kb). It also achieves the best performance in terms of dRMSE across all resolutions. While all methods

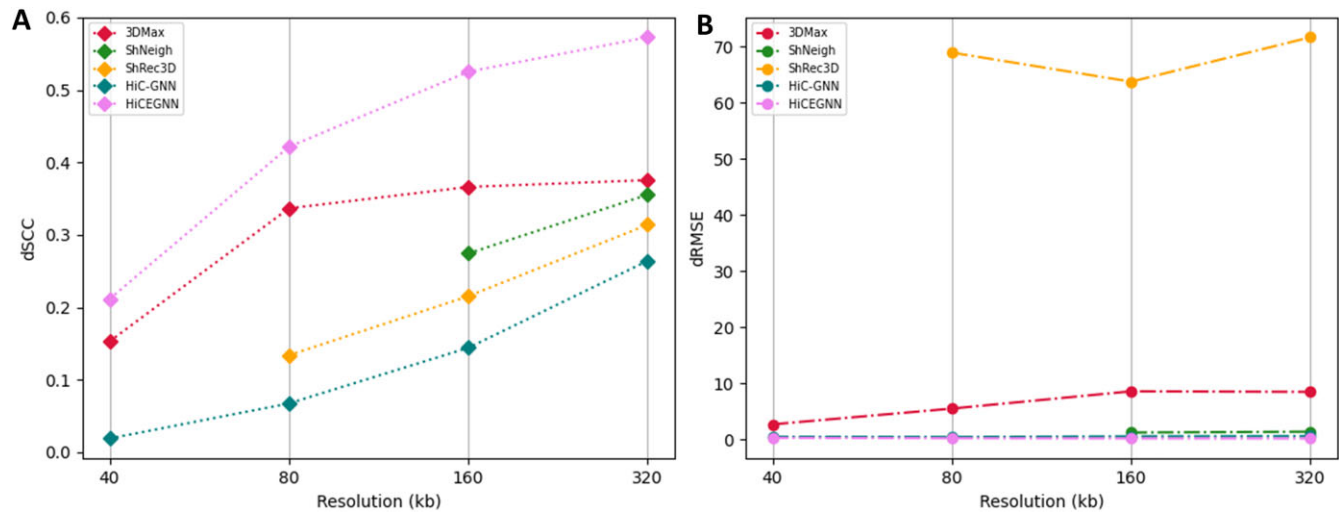


Figure 2. The average dSCC and dRMSE of the five methods on 3D chromosome structure construction from Schi-C test dataset *human_cell_1_test_data* across different resolutions (40 kb, 80 kb, 160 kb, and 320 kb). **(A)** The dSCC scores of the methods. **(B)** The dRMSE scores of the methods. If a method fails to generate chromosome structures at a resolution, its value at the resolution point is missing.

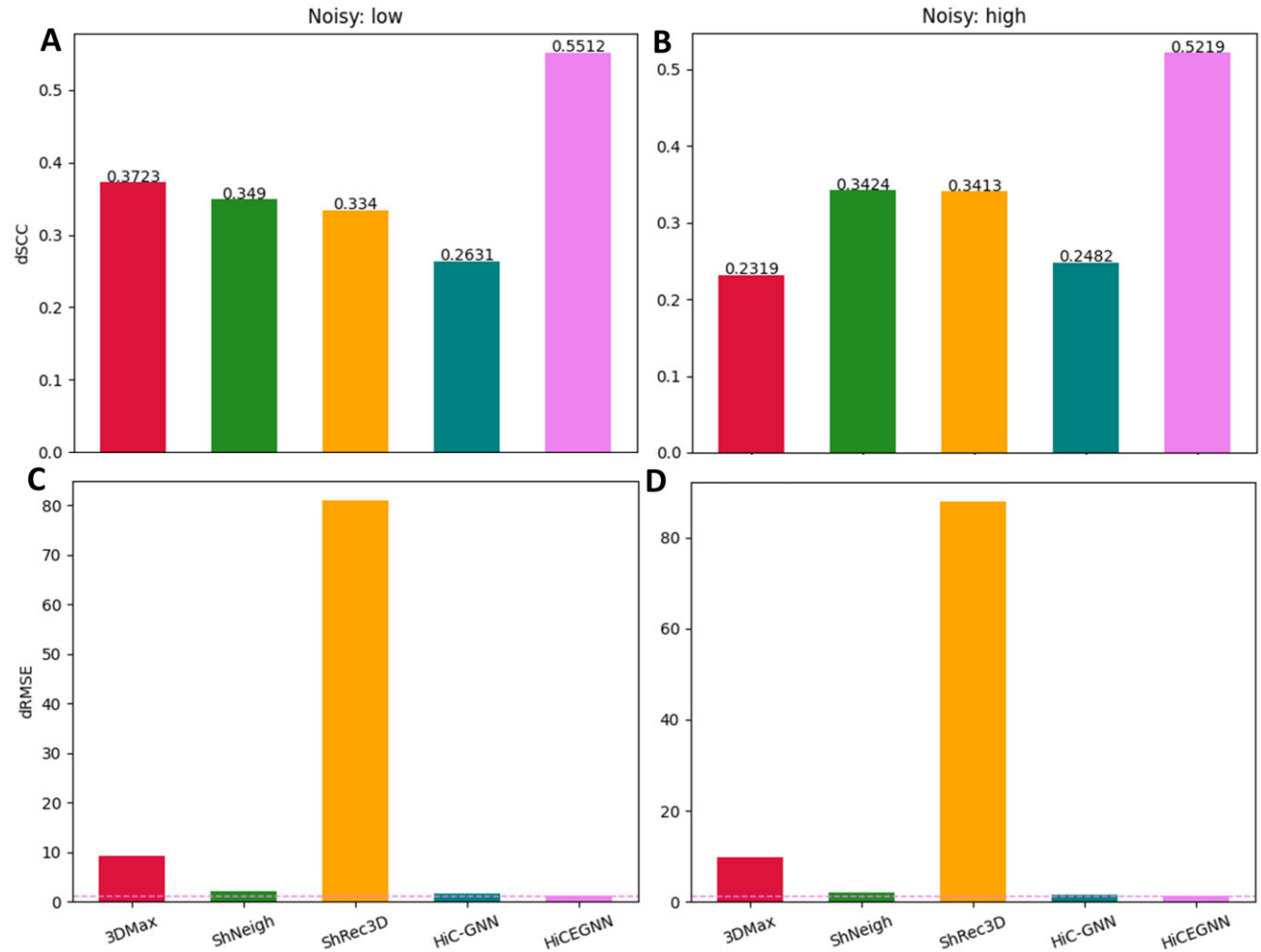


Figure 3. The average dSCC and dRMSE of the five methods on *human_cell_1_test_data* at two noise levels (high and low). **(A)** The dSCC scores at the low noise level. **(B)** The dSCC scores at the high noise level. **(C)** The dRMSE scores at the low noise level. **(D)** The dRMSE scores at the high noise level. The dashed line in panels (C) and (D) marks the lowest dRMSE value from HiCEGNN.

can construct 3D structures at low resolutions (e.g. 320 kb), some methods, such as ShNeigh and ShRec3D, fail to reconstruct 3D structures at high resolutions (e.g. 40 kb). Regarding dSCC, the performance of all methods increases as the resolution decreases from high resolution (e.g. 40 kb) to low resolution (e.g. 320 kb). This is expected because there are more points and less data per point at high resolution than at low resolution.

HiCEGNN is robust against noise in ScHi-C data

To evaluate how robust HiCEGNN is against the level of noise in ScHi-C data, we compare it with the other methods on ScHi-C test dataset *human_cell_1_test_data* at a low resolution of 320 kb across two noise levels (low and high). Here, 320 kb resolution is used because at this resolution all the five methods can generate 3D chromosome structures from the ScHi-C data.

As shown in Fig. 3A and B, HiCEGNN has the highest dSCC values 0.5512 and 0.5219, at both low and high noise levels, which are substantially higher than the dSCC values of the other methods. For all the methods, their dSCC score decreases when the noise level is increased from low to high as expected. However, HiCEGNN is rather insensitive to the change of noise levels as, its dSCC value only slightly drops from 0.5512 to 0.5219, while dSCC score of some method, such as 3DMax, substantially drops (e.g. from 0.3723 to 0.2319 for 3DMax). As shown in Fig. 3C and D, the dRMSE score of HiCEGNN is the lowest among all the methods at both low and high noise levels. A larger dSCC value or a smaller dRMSE value typically indicates that the predicted structure is closer to the ideal 3D structure (i.e. the desired outcome), and vice versa. For all the methods, including HiCEGNN, an increase in noise levels in ScHi-C data leads to smaller dSCC values and larger dRMSE values, demonstrating that higher noise levels degrade the quality of the reconstructed 3D structures.

Conclusion

In this work, we developed a new deep learning method, HiCEGNN, based on a GNN that is equivariant to the rotation and translation of 3D objects to generate 3D chromosome structures from ScHi-C data. It was trained to make the generated 3D chromosome structures match the chromosomal distances calculated from ScHi-C data as well as possible. Once trained, it can be directly applied to any Hi-C data to infer 3D chromosome structures quickly without the need to run an expensive optimization process to simulate chromosome structures as most traditional 3D chromosome structure modeling methods do. Moreover, because it is a machine learning approach, it benefits from learning from common patterns in multiple Hi-C training data, and its performance can continue to rise as more training data is used. This is different from the traditional nonmachine learning optimization/simulation methods that apply the same optimization process to each input dataset separately without learning from different datasets.

HiCEGNN not only outperforms the three selected traditional optimization methods and the only other deep learning method, but it also generalizes well across different cells within the same cell type (line), different cell types, and different species. Its performance is also robust against the increase

of the noise in ScHi-C data. It also works well on bulk Hi-C data of a population of cells, indicating it is a general, reliable approach to generating 3D genome structures from all kinds of Hi-C data.

In the future, we plan to train HiCEGNN on a large dataset consisting of ScHi-C data of multiple species to further enhance its modeling accuracy and generalization performance.

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Author contributions: J.C. conceived the project. J.C. and Y.W. designed the method. Y.W. implemented the method and collected the data. Y.W. and J.C. wrote the manuscript.

Supplementary data

Supplementary data is available at NAR Genomics & Bioinformatics online.

Conflict of interest

None declared.

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Data availability

The cooler file containing both the ScHi-C data of the three human cells (GEO accession number: GSE130711) and population human Hi-C data (GEO accession number: GSE130711) was downloaded from <https://salkinstitute.app.box.com/s/fp63a4j36m5k255dhje3zcj5kfuzkyj1>. The cooler file containing both the ScHi-C data of two *Drosophila* cells (GEO accession number: GSE131811) and population *Drosophila* Hi-C data (GEO accession number: GSE131811) was obtained from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE131811>. The training, validation, and test datasets generated from the downloaded Hi-C datasets are available at <https://doi.org/10.5281/zenodo.10976222>. The source code of HiCEGNN is available at FigShare: <https://doi.org/10.6084/m9.figshare.27119772>.

Code availability

The source code of HiCEGNN is available at the GitHub repository: <https://github.com/BioinfoMachineLearning/HiCEGNN> and at FigShare: <https://doi.org/10.6084/m9.figshare.27119772>. The LINE package used by HiCEGNN is available at <https://github.com/tangjianpku/LINE>.

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